```
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import (
    mean_absolute_error,
    mean_squared_error,
    r2_score
)
housing = fetch_california_housing(as_frame=True)

df = housing.frame

df.head()

df.info()

df.describe()

X = df.drop("MedHouseVal", axis=1)

y = df["MedHouseVal"]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)

model = LinearRegression()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

mae = mean_absolute_error(y_test, y_pred)

rmse = mean_squared_error(y_test, y_pred) ** 0.5

r2 = r2_score(y_test, y_pred)

print("MAE :", round(mae,3))
print("RMSE :", round(rmse,3))
print("R2 Score :", round(r2,3))

plt.figure(figsize=(6,4))

plt.scatter(y_test, y_pred)

plt.xlabel("Actual Price")

plt.ylabel("Predicted Price")

plt.title("Actual vs Predicted House Prices")

plt.grid(True)

plt.show()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20640 entries, 0 to 20639
Data columns (total 9 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   MedInc       20640 non-null  float64
 1   HouseAge     20640 non-null  float64
 2   AveRooms     20640 non-null  float64
 3   AveBedrms    20640 non-null  float64
 4   Population   20640 non-null  float64
 5   AveOccup     20640 non-null  float64
 6   Latitude     20640 non-null  float64
 7   Longitude    20640 non-null  float64
 8   MedHouseVal  20640 non-null  float64
dtypes: float64(9)
memory usage: 1.4 MB
MAE : 0.533
RMSE : 0.746
R2 Score : 0.576
```